

习题 5.4

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$$\frac{d}{dx}f(x^2) = f'(x) \cdot 2x$$

和

$$\frac{d^2}{dx^2}f^2(x) = C_2^0 f''(x)f(x) + C_2^1 f'(x)f'(x) + C_2^2 f(x)f''(x)$$

所以在 $x = 1$ 处，两个导数都等于 0。

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• (1)

$$\begin{aligned}\frac{dy}{dx} &= \frac{3a\sin^2(t)\cos(t)}{-3a\cos^2(t)\sin(t)} \\ &= \frac{\sin^2(t)\cos(t)}{-\cos^2(t)\sin(t)} \\ &= -\frac{\sin(t)}{\cos(t)}\end{aligned}$$

于是

$$\begin{aligned}\frac{d^2y}{dx^2} &= \frac{\left(-\frac{\sin(t)}{\cos(t)}\right)'}{(a\cos^3(t))'} \\ &= \frac{-\frac{1}{\cos^2(t)}}{-3a\cos^2(t)\sin(t)} \\ &= \frac{1}{3a\cos^4(t)\sin(t)}\end{aligned}$$